1 **(a)** The result from the 2000 experiment. It has the smallest range of uncertainty. (accept smallest uncertainty / error) [1]

(b) **(i)** An error which results in values being always higher or always lower than expected / true value. [1]

(ii) There might be systematic errors in this experiment [1]
which would shift the result away from the true value (without affecting the precision of the measurement.) [1]

(c) $F = GMm / R^2$
Making R the subject, $R = (GMm / F)^{½}$

Using 2009 results for G & fractional uncertainty method,
$\Delta R / R = ½ (0.00017 / 6.67349 + \Delta M / M + \Delta m / m + \Delta F / F)$ [1]
$\Delta R / (3.844 \times 10^8) = ½ (0.00017 / 6.67349 + 0.01 + 0.02 + 0.02 / 1.98)$

$\Delta R = 0.08 \times 10^8$ m (1 sf) [1]
$R \pm \Delta R = (3.84 \pm 0.08) \times 10^8$ (correct dp for R) [1]

2 (a) (i)